Runtime Assurance for Sampling-Based Optimizers in Flood Re-Planning: Exact Verification and Certified Escalation

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***Abstract*—Quantum and annealing-based optimizers are proposed for disaster logistics, but they return samples with no guarantee of validity, and their penalty encodings are often inexact. We study a re-planning loop that reassigns evacuation groups to shelters as roads close and shelters fail. The loop verifies every sample exactly, falls back to a greedy rule and, before handing a case to a human, asks a mixed-integer program (MILP) whether any valid plan exists. On 180 events of a flood digital twin (12 binary variables), dispatching the most frequent sample unchecked would have sent an invalid plan in 16.6% of draws on solvable events for the Quantum Approximate Optimization Algorithm (QAOA) and 6.5% for simulated annealing. Even the best penalty-free QAOA sample was invalid in 5.9% of solvable draws, because the pairwise capacity penalty misses overloads by three groups; the verified loop dispatched none. On 360 instances built on OpenStreetMap road networks of six Indian cities, with up to 320 variables, the annealer's valid-plan rate fell from 71.3% at 8 groups to 17.0% at 32, and an uncertified loop escalated 4.4% of solvable cases to a human. The MILP settled every instance, the slowest in 40.8 s, so a solver time-out must never be read as infeasibility. Calling the MILP first was faster in median time and always optimal; at these sizes the samplers add no value, and we claim no quantum advantage.**

***Index Terms*—runtime assurance, quantum optimization, simulated annealing, mixed-integer programming, evacuation planning**

I. Introduction

When a flood closes roads and fills shelters, the assignment of evacuation groups to shelters has to be re-made, often many times in one night. The assignment is a small constrained optimization problem, and it is often used as a test problem for quantum and quantum-inspired optimizers, which take it as a quadratic unconstrained binary optimization (QUBO) problem [1]–[4]. For an operations room, two properties of these tools matter more than speed. They return *samples*, so the same call can give a different, possibly invalid answer. And the QUBO they solve encodes capacity limits with penalties, which is exact only in special cases: a sample that satisfies every penalty can still overfill a shelter.

Safety-critical systems run unreliable components inside a *runtime-assurance* architecture, in which a simple verified controller checks or replaces the complex one [5], [6], as shields do for learned policies [7]. For a sampling optimizer we ask how often its output would be unsafe without the check, how the loop should decide between trying harder and calling a human when no sample is valid, and how many samples it should draw. Our contributions, for flood re-planning, are:

1) A runtime-assurance loop for sampling optimizers with an exact plan verifier, a greedy fallback and a *certified escalation* step that uses a mixed-integer program (MILP) to decide whether any valid plan exists (Fig. 1).
2) A measurement on twin episodes with QAOA [8], [9] and simulated annealing [10] of how often unverified dispatch would be unsafe, for the most frequent, the lowest-energy and the best penalty-free sample, with a classification of every unsafe sample by the rule it breaks.
3) A city-scale study on OpenStreetMap road graphs of six Indian cities, beyond the reach of enumeration, showing where samplers and uncertified loops fail and what the certificate costs in time.
4) A comparison with an exact-solver-first loop in time and plan cost, and a shot-budget bound with an adaptive budget.

All QAOA results are classical simulations; we make no claim of quantum speed-up.

TABLE I
POSITION OF THIS WORK

Work	Unreliable part	Check	Escalation
Simplex, SOTER [5], [6]	controller	envelope	safe controller
Shielding [7]	RL policy	automaton	blocked action
QUBO for disasters [3]	annealer	–	–
This work	sampler	exact verifier	MILP-certified

II. Related Work

Runtime assurance. The Simplex architecture switches from an unverified high-performance controller to a verified safety controller when the former leaves a safe envelope [5]; SOTER brings the pattern to robotics software [6], shields apply it to reinforcement learning [7], and runtime verification checks executions against a specification as they happen [11]. Levels-of-automation theory argues that automation should fall as the cost of an error rises [12]. Our verifier is the switching logic, and the MILP plays the role of the verified controller of last resort.

Samplers as tools. Variational quantum algorithms are hybrid loops in which a classical optimizer calls a quantum circuit [13]; QAOA with constraint-preserving mixers keeps one-hot structure but still penalizes capacity [9], [14]. Quantum annealing has been applied to disaster resource allocation [3], and reviews stress that practical advantage in optimization is unproven [4]. Agent frameworks let language models call tools [15], [16], and runtime enforcement constrains the actions such agents may take [17]; these works do not check the plans an optimizer returns.

Evacuation planning. Shelter allocation has long been solved with integer programming [18]; modern MILP solvers such as HiGHS [19] make exact solutions cheap at the sizes of a city district. Repairing a plan rather than re-planning from scratch reduces disruption [20].

Table I summarizes the gap. We found no application of runtime assurance to sampling optimizers, and the sampler studies cited here report solution quality per call; none measures what happens when a loop acts on the answer.

III. Problem and Loop

A. Assignment

Groups $g = 1, \dots, G$ of size n_g are assigned to shelters $s = 1, \dots, S$ with remaining seats R_s. With travel cost c_{gs} on the best open path and a set $\mathcal{F}$ of forbidden pairs (no open path, failed shelter, group larger than the seats left), a plan $\mathbf{x} \in \{0,1\}^{G \times S}$ is valid when

$$V(\mathbf{x}) = \Big[\textstyle\sum_s x_{gs} = 1 \ \forall g\Big] \wedge \Big[x_{gs} = 0 \ \forall (g,s) \in \mathcal{F}\Big] \wedge \Big[\textstyle\sum_g n_g x_{gs} \le R_s \ \forall s\Big], \quad (1)$$

and the planner minimizes $\sum_{g,s} c_{gs} x_{gs}$ over valid plans. Samplers solve the QUBO with a per-group shifted cost, a one-hot penalty and a pairwise penalty $P\, x_{gs} x_{hs}$ for every two groups that together exceed R_s [2]. The pairwise term misses overloads by three or more groups whose pairs all fit, so a sample can satisfy every penalty and still violate (1). Exact encodings exist, with slack variables [2] or one penalty term per minimal overloading set of groups, at the cost of extra qubits or higher-order terms; we keep the pairwise form, which needs neither, and treat its blind spot as one of the failures the loop must catch.

Algorithm 1 Runtime-assurance re-planning at one event

```
draw shots in batches of b until a sample passes V or the budget N is used
x ← cheapest sample with V(x); if none, x ← greedy
if V(x) then
    dispatch x
else
    solve (2) with a time limit
    if feasible, dispatch its plan; if infeasible, escalate with the certificate
    if timed out, escalate marked as uncertified
end if
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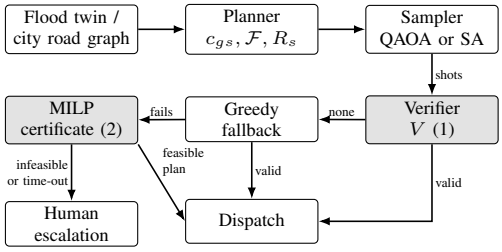


Fig. 1. Runtime-assurance loop. Shaded blocks are exact, and the sampler is never trusted. A human is called when the MILP proves that no valid plan exists, or, flagged as uncertified, when it runs out of time.

B. Runtime-Assurance Loop

Algorithm 1 gives the loop. It verifies every sample with (1) against the state at dispatch time, keeps the cheapest valid one, falls back to a greedy rule (largest group first to its cheapest shelter with seats left), and only then asks the MILP

$$\min_{\mathbf{x}} \sum_{g,s} c_{gs} x_{gs} \quad \text{s.t. (1)}, \ x_{gs} \in \{0,1\}. \quad (2)$$

If (2) is infeasible, escalation to a human is *certified*: no valid plan exists, within the solver's floating-point feasibility tolerance (optimality holds within its default MIP gap). If it is feasible, the loop dispatches its plan instead of escalating. If the solver stops at its time limit, the loop escalates but marks the case *uncertified*, so the officer knows that a plan may exist. A loop that escalates whenever the sampler and greedy rule fail, without asking the MILP, is the natural baseline; we call it the uncertified loop.

C. Shot Budget

If one shot is valid with probability q, the chance that N independent shots contain no valid plan is $(1-q)^N$, so

$$N \ \ge \ \frac{\ln \epsilon}{\ln(1-q)} \quad (3)$$

shots make that chance at most ϵ. Since q is unknown and changes with every event, the loop draws batches of $b = 8$ shots and stops at the first batch with a valid sample, up to $N = 64$. A wrong guess of q then costs shots or plan quality, while the verifier and certificate still stand behind every decision.

IV. Experimental Setup

Twin episodes. We replay 30 seeded episodes of a flood digital twin with four groups and three shelters (12 binary variables): the initial state, closure of road R5, a shelter surge of half its capacity, a further random closure, the failure of shelter A and, as a final stress, the failure of shelter B (180 re-planning events, 27 of them without any valid plan). QAOA uses a ring-XY mixer at depth 3, trained per event by COBYLA with layer-wise interpolation [21]. Simulated annealing runs on the same QUBO as 64 independent chains per call, one shot per chain, with 30 sweeps. Each event is sampled ten times with independent seeds, 64 shots per draw.

City instances. We built 360 instances on the OpenStreetMap road graphs of Delhi, Chennai, Mumbai, Kolkata, Guwahati and Patna, with the mapped zones and ten shelters per city, $G \in \{8, 16, 32\}$ groups (80–320 variables), demand of 60% or 90% of the seats left, and a disruption that closes the three most flood-exposed roads and fails one shelter. Travel times follow the prototype's router (flood- and traffic-weighted length at 25 km/h); capacities are building-type estimates from the city data, and group sizes are synthetic. Simulated annealing runs 64 chains of 40 sweeps per call, ten times per instance; on three instances per city, size and demand we repeated it with 400 sweeps. At this scale the annealer is the only sampler; QAOA is not simulated. All city results come from one run on a frozen copy of the city data, whose SHA-256 sums are released with the scripts. Enumeration of 10^G plans is impossible here, so the MILP is the only exact reference.

Metrics. Unsafe dispatch: the sample a naive loop would act on violates (1). We score three such integrations: the most frequent sample, the lowest-energy sample, and the lowest-energy sample among those with zero penalty, i.e. the samples that pass the encoding's own constraints. False escalation: a case escalated although (2) has a solution. Optimality gap: excess travel cost of the dispatched plan over the MILP optimum. Times are wall-clock on a laptop running three instances in parallel. The draws of one event share its instance, so they are not independent samples of the event population.

V. Results

A. Twin Episodes

Table II gives the twin results over 180 events and ten draws each. Dispatching the most frequent sample unchecked would have been unsafe in 29.1% of event draws for QAOA and 20.6% for simulated annealing. These include the 27 infeasible events, where any answer a sampler gives is wrong. The verified loop dispatched no invalid plan, escalated every infeasible event and no feasible one, and dispatched an optimal plan in 93.1% (QAOA) and 95.4% (annealing) of feasible draws. At this size the uncertified loop also behaved well (0.0% and 0.0% false escalations), because the greedy rule found a plan whenever the sampler did not. The MILP took a median of 6.9 ms against 1.4 ms for full enumeration, while one QAOA training took 13.0 s of simulation. Resampling events gives 95% intervals of 22.7–35.6% for QAOA and 14.3–26.4% for annealing. On the 153 solvable events, the top QAOA sample was unsafe more often than the top annealing sample in 24 events and less often in 7 (Wilcoxon signed-rank $p = 2.4 \times 10^{-4}$), so the difference is consistent across events and not an effect of a few draws.

TABLE II
Twin episodes: 180 events × 10 draws, 12 variables

Sampler, budget	Unsafe if unverified	Valid sample	Optimal dispatched	Shots used	False escal.
QAOA, 64 shots	29.1	97.1	93.1	64.0	0.0
QAOA, adaptive	28.8	97.1	68.8	18.0	0.0
SA, 64 shots	20.6	95.4	95.4	64.0	0.0
SA, adaptive	21.0	95.4	95.4	18.7	0.0

All in % except shots. False escal.: uncertified loop; the certified loop had none.

TABLE III
One twin episode with QAOA (seed 178, 10 draws per event)

Event	Plan exists	P(valid) of state	Top sample unsafe	Plan source (draws)	Max. gap %
E0 initial	yes	0.98	0/10	sampler 10	0.0
E1 R5 closed	yes	0.98	0/10	sampler 10	0.0
E2 surge	yes	0.02	10/10	sampler 7, greedy 3	1.0
E3 closure	yes	0.00	10/10	greedy 10	5.3
E4 A failed	yes	0.00	10/10	greedy 8, sampler 2	5.3
E5 B failed	no	0.00	10/10	escalate 10	–

Table III follows one episode with QAOA through its six events (seed 178, the one in which the greedy rule did the most work). In the first two events the trained state put almost all of its probability on valid plans. After the surge it put only 0.02 there, and after the second closure 0.00, so the most frequent sample was invalid in every draw from then on. After the surge the verifier still picked out a rare valid sample in some draws; otherwise the greedy rule supplied the plan, and no dispatched plan cost more than 5.3% above the optimum. When shelter B failed, the MILP proved that no plan existed and the loop escalated with a certificate.

Table IV splits the twin results by event. In the initial state and after the first road closure the most frequent sample was almost always valid. The risk appeared once capacity became tight: after the shelter surge, a second closure and the failure of shelter A, the unsafe rate of QAOA rose from 23.7% to 33.0%, and that of annealing from 7.7% to 13.3%. After shelter B also failed, 27 of 30 events had no valid plan at all. Counting solvable events only, unverified dispatch would still have been unsafe in 16.6% (QAOA) and 6.5% (annealing) of draws.

TABLE IV
TWIN RESULTS BY EVENT: UNSAFE IF DISPATCHED UNVERIFIED, OPTIMAL UNDER THE VERIFIED LOOP (30 EVENTS EACH; % OF DRAWS)

Event	No valid plan	Unsafe		Optimal	
		QAOA	SA	QAOA	SA
E0 initial	0/30	0.0	1.0	100.0	100.0
E1 R5 closed	0/30	0.0	1.0	100.0	100.0
E2 surge	0/30	23.7	7.7	85.0	93.3
E3 closure	0/30	28.0	10.3	90.3	93.3
E4 A failed	0/30	33.0	13.3	89.3	90.0
E5 B failed	27/30	90.0	90.0	100.0	100.0

TABLE V
CITY INSTANCES ON OPENSTREETMAP ROAD GRAPHS (6 CITIES)

G	vars	demand	feasible inst.	valid sample %	unsafe %	false esc. uncert. %	MILP s (max)
8	80	60%	53/60	68.3	55.7	0.0	0.73
8	80	90%	9/60	88.9	86.7	11.1	0.64
16	160	60%	51/60	39.4	85.0	0.0	1.33
16	160	90%	44/60	28.9	82.7	23.6	5.39
32	320	60%	50/60	18.0	85.0	0.0	13.88
32	320	90%	50/60	16.0	86.7	0.0	40.76

valid: verified plan in 64 SA shots; unsafe: top sample unchecked. No run reached the 60 s limit.

B. City Scale

Table V and Fig. 2 give the city results. As the problem grew from 8 to 32 groups, simulated annealing produced a valid plan in 71.3% and then 17.0% of calls on solvable instances; Section V-C shows that at this scale most samples were not even complete assignments. Dispatching the top sample unchecked would have been unsafe in most calls at every size. With 8 groups at 90% demand each group is about as large as the free seats of an average shelter, and only 9 of 60 instances had a valid plan. The uncertified loop escalated 4.4% of solvable cases overall (11.1% at 90% demand, 0.0% at 60%), because the greedy rule also fails when capacity is tight. Where the greedy rule did find a plan, its median cost was 7.6% above the optimum. Within the 60 s limit the MILP settled 360 of the 360 instances, finding the optimal plan or proving that none exists, in at most 40.8 s. No solve reached the 60 s limit; the slowest took 40.8 s. A 5 s deadline, short enough to re-plan within one radio cycle, would have stopped 7 instances with neither a plan nor a proof, and every one of them had a valid plan. A time-out must therefore be reported as unknown, never as infeasible.

Fig. 3 shows what the certificate costs. The median MILP time rose from 0.01 s at 80 variables to 0.32 s at 320, so at district scale a typical certificate takes under a second; the cost lies in the rare slow instances. Resampling the solvable instances puts the false-escalation rate of the uncertified loop at 2.3–6.9% (95% interval). The certified loop never escalated a solvable case, but its plans were often not optimal: over 2,570 calls on solvable instances it dispatched an optimal plan in 37.0%, with a median gap of 4.4% and a 90th percentile of 28.7%, because the greedy rule supplied 1516 plans, the sampler 940 and the MILP 114.

Table VI gives the same results per city. Unverified dispatch was unsafe in 12.7 to 100.0% of calls, depending on the city. In Kolkata the three closures cut one of the six zones off from every shelter, so any instance with a group there has no valid plan, and only 5 of 60 were solvable. Among the other five cities, Chennai had the most false escalations (8.0%) and the weakest greedy rule, which found a plan in 92.0% of its solvable instances. The failures of the uncertified loop therefore depend on the city network as well as on the problem size.

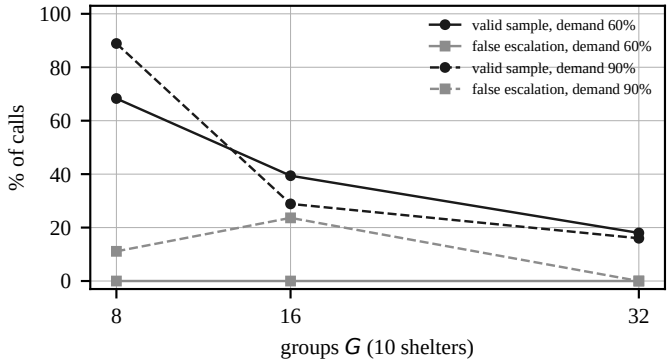


Fig. 2. City scale: share of calls in which the sampler produced a valid plan, and share of solvable cases an uncertified loop would escalate, against the number of groups.

C. Why Unverified Samples Fail

Table VII classifies every unsafe top sample by the first rule of (1) it breaks. On the twin, 97.7% of the unsafe QAOA samples and 94.6% of the annealing samples overfilled a shelter although every pair of groups in it fitted, so the QUBO assigned them no penalty at all. Overloads with a penalized pair almost never reached the top, and the XY mixer never broke the one-hot rule. At city scale the annealer, run for a fixed 40 sweeps, left groups with no shelter or with several in 34.5% of unsafe samples at 8 groups, 86.6% at 16 and all of them at 32, while penalty-free overloads fell from 62.4% to none. On the twin the failures are encoding misses; at city scale they are mostly broken one-hot rows, which longer annealing did not repair (Section V-D). The verifier catches both. The twin re-run that produced these classes reproduced the original flags in every draw; the city classes come from the same run as the city tables.

D. Usual Practice and an Exact Solver First

Table VIII scores the integrations that sampler studies usually rely on. Taking the lowest-energy sample instead of the most frequent one lowered QAOA's unsafe rate on solvable twin events from 16.6% to 5.9%, and keeping only samples with zero penalty gave 5.9%; for annealing the three rates were 6.5, 5.9 and 5.9%. The remaining failures are the penalty-free overloads of Section V-C: they carry no penalty and can cost less than any valid plan, so no choice by energy can exclude

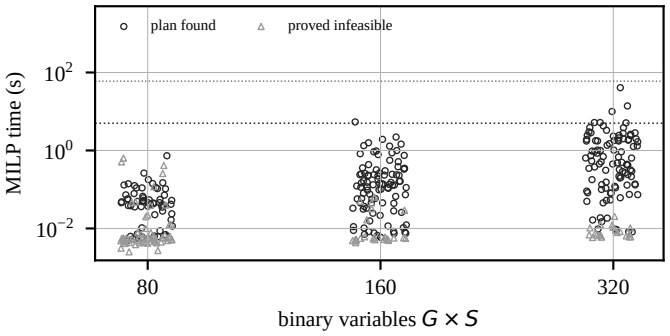


Fig. 3. Cost of the certificate: MILP solve time of every city instance (log scales, sizes jittered). The dotted line marks a 5 s deadline; every instance above it had a valid plan.

TABLE VI
CITY INSTANCES BY CITY (60 INSTANCES EACH; % UNLESS NOTED)

City	Solvable	Valid sample	Unsafe	False esc. uncert.	Greedy found	MILP s (median)
Guwahati	58/60	93.1	12.7	0.0	100.0	0.01
Patna	50/60	5.2	100.0	6.0	94.0	0.23
Mumbai	50/60	21.6	96.7	3.8	96.0	0.15
Chennai	50/60	22.6	86.5	8.0	92.0	0.19
Delhi	44/60	30.2	85.8	3.4	95.5	0.06
Kolkata	5/60	40.0	100.0	20.0	80.0	0.01

them. At city scale the lowest-energy sample was unsafe in 71.2% of calls on solvable instances, and in 14.0% of calls no sample was penalty-free at all.

We next compare sampler-first and solver-first loops. In a separate timing run on the twin the MILP took a median 5.1 ms per event, one annealing call of 64 chains 32 ms and one QAOA training 9.3 s of simulation. At city scale one annealing call took a median 0.32, 0.84 and 1.94 s at 8, 16 and 32 groups, against 0.01, 0.08 and 0.32 s for the MILP. The MILP was slower than one annealing call on 39 solved instances (20% of those with 32 groups), and the annealer found a valid plan on 1 of them. An exact-solver-first loop would therefore have been faster in median time and optimal, whereas the verified loop fell back to the greedy plan whenever the sampler failed, at a median 7.6% above the optimum at city scale. More search did not help. On 108 city instances (three per city, size and demand) an annealer with 400 instead of 40 sweeps found a valid plan in 70.2, 35.6 and 13.3% of calls on solvable instances at 8, 16 and 32 groups, against 74.2, 38.6 and 13.3% with 40 sweeps on the same instances; per instance it did better on 1 and worse on 5, at 3.8, 9.1 and 21.2 s per call. The limit is the encoding and its penalty weight, not the length of the search.

E. Shot Budget

With the adaptive budget the twin loop used 18.4 shots per event on average instead of 64 and took the same escalation decision in every draw. Plan quality did change for QAOA: it dispatched an optimal plan in 68.8% of feasible draws against 93.1% with the full budget, because the loop keeps the cheapest valid sample of the first successful batch rather than of all 64 shots. Annealing dispatched an optimal plan whenever it found a valid one, with either budget, so its rate stayed at 95.4%. At city scale the adaptive rule used 50.0 shots per call, more than on the twin, because valid samples are rarer there. By (3), a sampler with $q = 0.05$ needs 90 shots for $\epsilon = 0.01$, more than the 64 used here.

TABLE VII
FIRST RULE BROKEN BY THE UNSAFE TOP SAMPLES (% OF UNSAFE SAMPLES)

Sampler	Unsafe draws	One-hot broken	Forbidden pair	Overload penalized	Overload penalty-free
Twin, QAOA	524	0.0	1.9	0.4	97.7
Twin, SA	370	5.4	0.0	0.0	94.6
City, SA, $G = 8$	854	34.5	3.0	0.0	62.4
City, SA, $G = 16$	1006	86.6	0.0	0.0	13.4
City, SA, $G = 32$	1030	100.0	0.0	0.0	0.0

TABLE VIII
UNSAFE IF DISPATCHED UNVERIFIED, BY INTEGRATION (% OF DRAWS)

Dispatched sample	Twin, QAOA all	Twin, QAOA solvable	Twin, SA all	Twin, SA solvable	City, SA solvable
most frequent sample	29.1	16.6	20.6	6.5	72.4
lowest-energy sample	20.1	5.9	20.0	5.9	71.2
lowest-energy penalty-free sample	19.5	5.9	19.4	5.9	57.2
verified loop (this work)	0.0	0.0	0.0	0.0	0.0

VI. DISCUSSION AND LIMITATIONS

Whether a solver gives good answers and whether a system built on it acts safely are separate questions. The second holds by construction whenever every sample is verified, a classical plan is always available and the certificate finishes in time, and it holds for any sampler plugged in. The sampler decides how often the exact components have to step in. At city scale the pairwise capacity QUBO broke down and the sampler's valid-plan rate fell to 17.0% at 32 groups, so an uncertified loop would have woken an officer for cases that a MILP solved in seconds. At every size we tested, calling the MILP first would have been faster in median time and always optimal; the samplers served only as stand-ins for future hardware, and the loop applies unchanged to any sampler.

The certificate has its own failure mode, the time limit. No run reached our 60 s limit, but deadlines of 1, 2, 5, 10 and 30 s would have stopped 45, 22, 7, 2 and 1 city instances with neither plan nor proof, and all of them had plans. An implementation that reads a time-out as "no plan" would have escalated them with a false certificate.

The results translate into four rules for anyone wiring a sampler into an operations loop:

1) Verify every sample against the state at dispatch time. Even on solvable twin events the top sample was unsafe in 6.5–16.6% of draws, and the best penalty-free sample in 5.9%; at city scale most calls were unsafe.
2) Ask an exact solver before calling a human, and call it first whenever it fits the deadline. Without it, 4.4% of

solvable city cases reached an officer, and the greedy fallback cost a median 7.6% more than the optimum.

3) Set the solver limit from the decision deadline and report a time-out as unknown, never as infeasible.
4) Keep the full shot budget when plan quality matters. The adaptive budget never changed an escalation, but it lowered QAOA's rate of optimal plans from 93.1% to 68.8%.

Threats to validity. The penalty weight P is set once per instance from the cost range, as in the usual QUBO recipe [2], and was not tuned; a larger P changes how often samples break the penalized rules, while the overloads that the pairwise term misses stay invisible at any weight; an exact encoding would remove those misses but not the broken one-hot rows of the city runs. QAOA was trained per event with a fixed budget and simulated without noise at 12 variables, and the city study uses simulated annealing as its only sampler, so we expect a noisy hardware sampler to produce valid samples less often. The draws of one event share its instance, which is why the intervals above resample events and instances rather than draws.

City demand is synthetic, capacities are estimates by building type, and every city instance uses the same kind of disruption: three closed roads and one failed shelter. The MILP is exact, but the slowest 320-variable instance needed 40.8 s, and regional problems will hit any fixed limit more often. Escalation was tested for infeasibility caused by failures and closures, and no officer took part in the evaluation, so the cost of a false escalation to a real operations room is not measured here.

VII. Conclusion

Sampling optimizers can join a flood re-planning loop safely if the loop never trusts them. An exact verifier checks every sample, a greedy rule and a MILP stand behind the sampler, and a human is called when the MILP proves that no plan exists or runs out of time, the second case flagged as uncertified. Without verification, the most frequent sample would have been unsafe in 16.6% of draws on solvable twin events for QAOA and 6.5% for annealing, and choosing the lowest-energy or the best penalty-free sample did not remove the risk, because the encoding cannot see some overloads. Without certification, the loop escalated 4.4% of solvable city cases. With both, no invalid plan was dispatched, and the MILP settled 360 of 360 city instances within 40.8 s. At these sizes an exact solver alone was faster and optimal, so the samplers add no value yet; we claim no quantum advantage.

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